

Trial Higher School Certificate Examination

2012



# Mathematics

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Begin each question in a new booklet
- Write your student number on each booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper. Detach.
- Multiple choice Answer sheet is at the back of this paper. Detach.
- Show all necessary working in Questions 11 – 16.
- Diagrams are not to scale.
- The mark allocated for each question is listed at the side of the question.

## Total Marks – 100

### Section I – Pages 2 – 4 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

### Section II – Pages 5 – 12 90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section

**Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.**

Section I – (10 marks)

Marks

Answer this section on the answer sheet provided at the back of this paper.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

1. The angle which the straight line  $3x + 5y + 2 = 0$  makes with the positive direction of the  $x$ -axis is closest to:

A.  $31^\circ$                       B.  $59^\circ$                       C.  $121^\circ$                       D.  $149^\circ$

2. Janet works out the sum of  $n$  terms of a given arithmetic series. Her answer, which is correct, could be:

A.  $S_n = 2(2^n - 1)$

B.  $S_n = 9 - 2n$

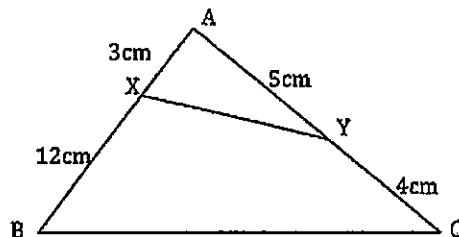
C.  $S_n = 8n - n^2$

D.  $S_n = 7 \times 2^{n-1}$

3. The values of  $x$  for which  $y = 2x^3 - 12x^2 + 18x + 7$  is increasing are:

A.  $x < 2$                       B.  $x > 2$                       C.  $1 < x < 3$                       D.  $x < 1$  or  $x > 3$

4.



If  $\triangle ABC$  has area  $36 \text{ cm}^2$  then the area of  $\triangle AXY$  is:

A.  $4 \text{ cm}^2$                       B.  $8 \text{ cm}^2$                       C.  $12 \text{ cm}^2$                       D.  $16 \text{ cm}^2$

5. When the curve of equation  $y = e^x$  is rotated about the  $x$ -axis between  $x = -2$  and  $x = 2$  the volume of the solid generated is given by:

A.  $\pi \int_{-2}^2 e^x dx$

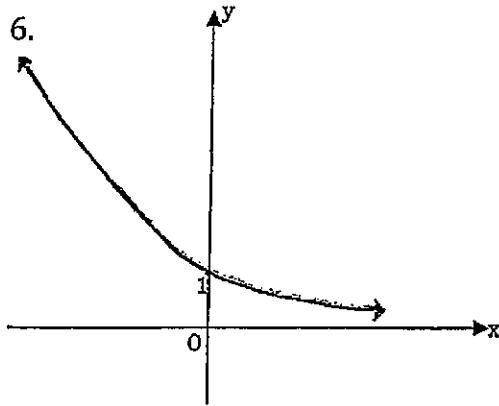
B.  $2\pi \int_0^2 e^{x^2} dx$

C.  $\pi \int_{-2}^2 e^{x^2} dx$

D.  $\pi \int_{-2}^2 e^{2x} dx$

Section I (cont'd)

Marks



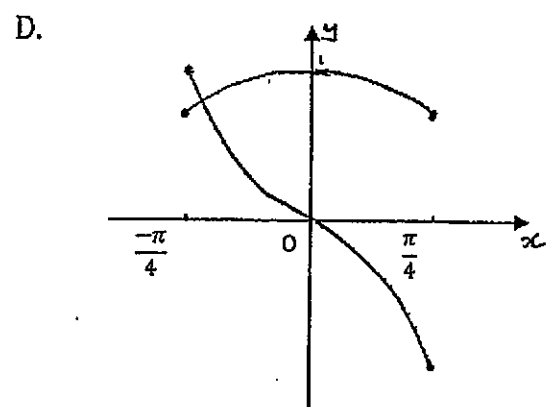
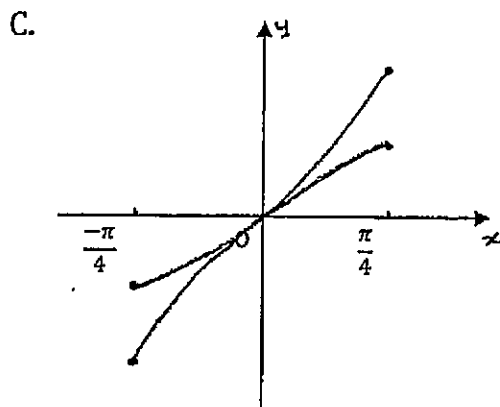
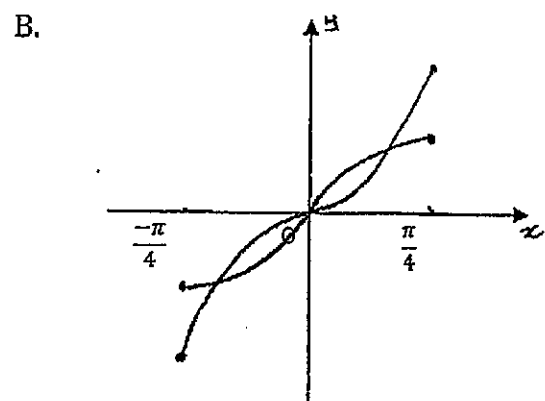
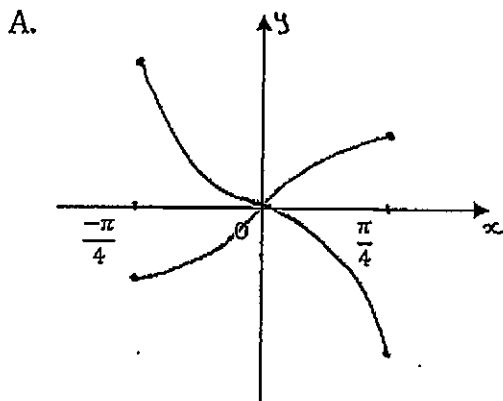
The graph illustrated could be:

- A.  $y = 2^x$
- B.  $y = (-2)^x$
- C.  $y = \left(\frac{1}{2}\right)^x$
- D.  $y = \left(-\frac{1}{2}\right)^x$

7. The quadratic function,  $Q(x) = 5x^2 - 4x + 3$ , has roots for  $Q(x) = 0$  of  $\alpha$  and  $\beta$ . Hence,  $\alpha^2 + \beta^2 =$

- A.  $\frac{46}{25}$
- B.  $\frac{29}{25}$
- C.  $\frac{-11}{25}$
- D.  $\frac{-14}{25}$

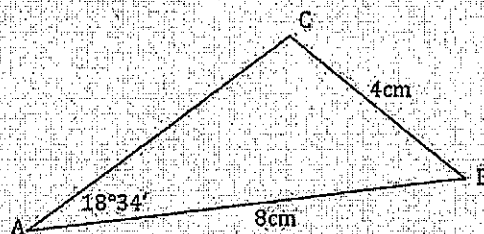
8. The graphs of  $y = \sin x$  and  $y = \tan x$  for  $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$  are represented in:



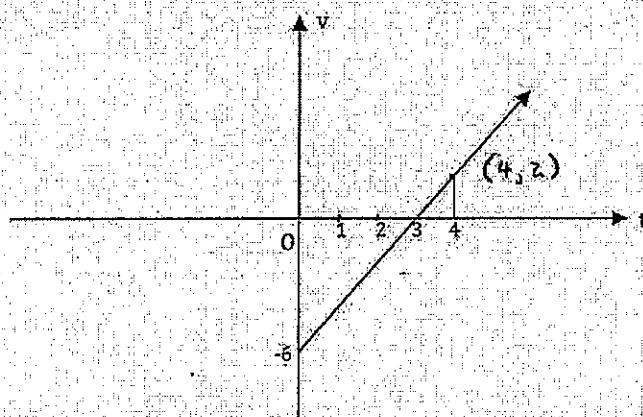
Section I (cont'd)

Marks

9. A possible answer to the size of  $\angle C$  in the triangle below is:



- A.  $140^{\circ}27'$       B.  $0^{\circ}10'$       C.  $37^{\circ}8'$       D. None of these answers
- 10.



The graph shows velocity expressed as a function of time. The distance travelled by the particle in the first 4 seconds is:

- A. 8 units      B. 10 units      C.  $4\sqrt{5}$  units      D. 12 units

Section II – Show all working

Question 11 – Start A New Booklet – (15 marks).

Marks

- a) Write the answer to  $\sqrt{\frac{4.83 \times 10.86}{17.83 - 5.92}}$  correct to 3 significant figures. 2
- b) Solve  $|2x - 3| \leq 5$  2
- c) If  $\log_a 2 = 0.36$  and  $\log_a 5 = 0.83$  evaluate  $\log_a \sqrt{10}$  2
- d) Differentiate each of the following with respect to  $x$
- (i)  $\cos 7x$  1
- (ii)  $\sqrt{e^{2x} + 4}$  2
- (iii)  $x \ln x$  2
- e) Find:
- (i)  $\int (3 - 2x)^4 dx$  1
- (ii)  $\int \frac{1}{\sqrt{x}} dx$  1
- (iii)  $\int \cos x^\circ dx$  2

Question 12 – Start A New Booklet – (15 marks)

Marks

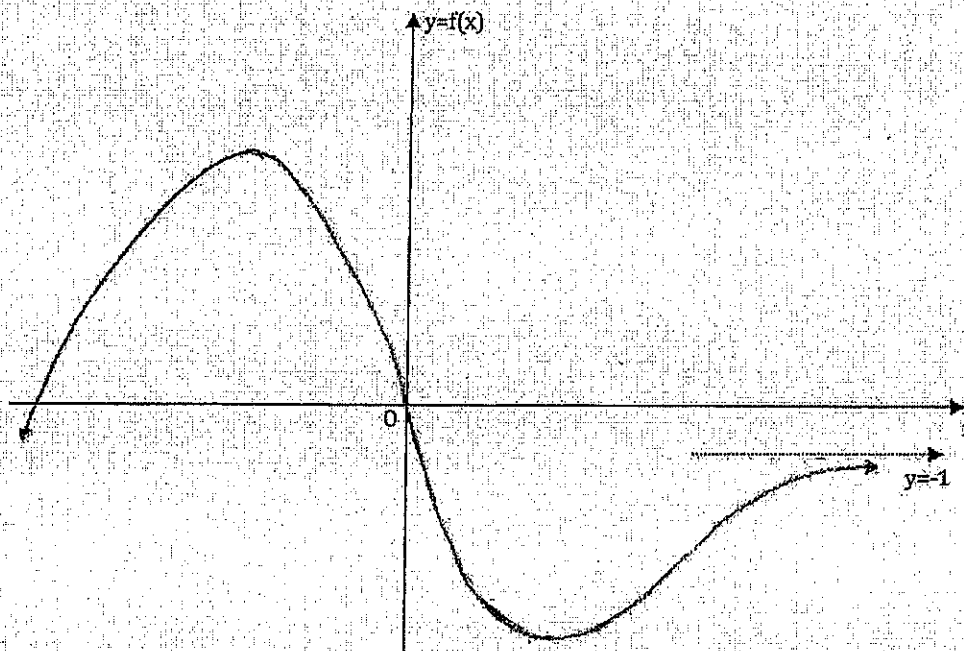
- a) Graph the region on the number plane given by  $y > \log_e(x - 1)$

2

- b) Copy this graph carefully onto your own paper. The graph shows  $y = f(x)$ .

On your graph draw the graph of  $y = f'(x)$  making it clear which graph is your answer.

2



- c) Initially a particle, travelling in straight line, is at rest at the origin. It is given an acceleration of  $(6t + 4)$  cm/sec<sup>2</sup>.

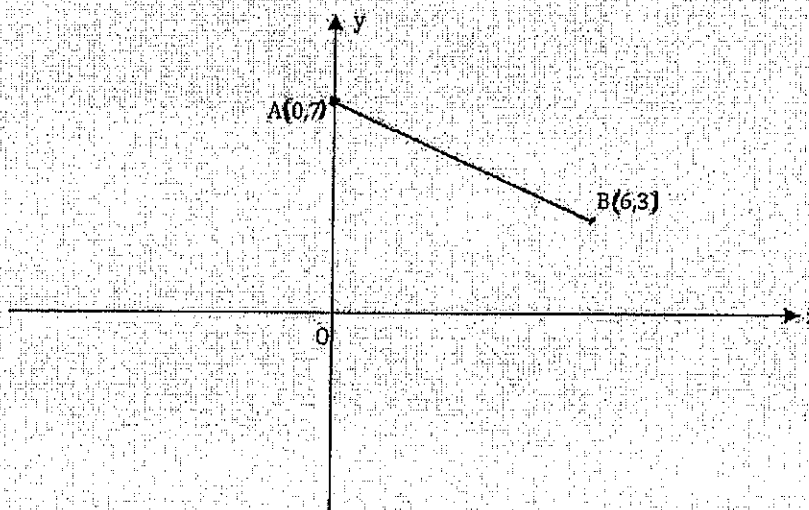
3

Find the motion equation for displacement.

Question 12 - (cont'd)

Marks

d)



$A(0, 7)$  and  $B(6, 3)$  are points on the number plane and the equation of  $AB$  is  $2x + 3y - 21 = 0$

- |                                                                                                |   |
|------------------------------------------------------------------------------------------------|---|
| (i) Find the length of $AB$                                                                    | 1 |
| (ii) Find the gradient of $AB$                                                                 | 1 |
| (iii) Show that the equation of the perpendicular from $D(-2, 0)$ to $AB$ is $3x - 2y + 6 = 0$ | 2 |
| (iv) Find the perpendicular distance from $D$ to $AB$                                          | 2 |
| (v) Find the coordinates of $C$ such that $ABCD$ is a parallelogram.                           | 1 |
| (vi) Find the area of parallelogram $ABCD$ .                                                   | 1 |

Question 13 – Start A New Booklet – (15 marks)

Marks

- a)  $20 + 10 + 5 + \dots$  is a geometric series. Find which term of the series will be just less than 0.0001.

3

- b) If  $\cos \theta = \frac{-8}{17}$  and  $\tan \theta < 0$ , find the exact value for  $\sin \theta$ .

2

- c) Sketch the graph of  $y = -3 \sin 2x$  for  $0 \leq x \leq 2\pi$

3

- d) Copy the table of values into your writing booklet and supply the missing numbers, for  $f(x) = x \sin x$ , writing each correct to 3 decimal places.

3

| $x$               | 1     | 1.5 | 2 | 2.5 | 3 |
|-------------------|-------|-----|---|-----|---|
| $f(x) = x \sin x$ | 0.841 |     |   |     |   |

Use Simpson's Rule with 5 function values to find an approximation for

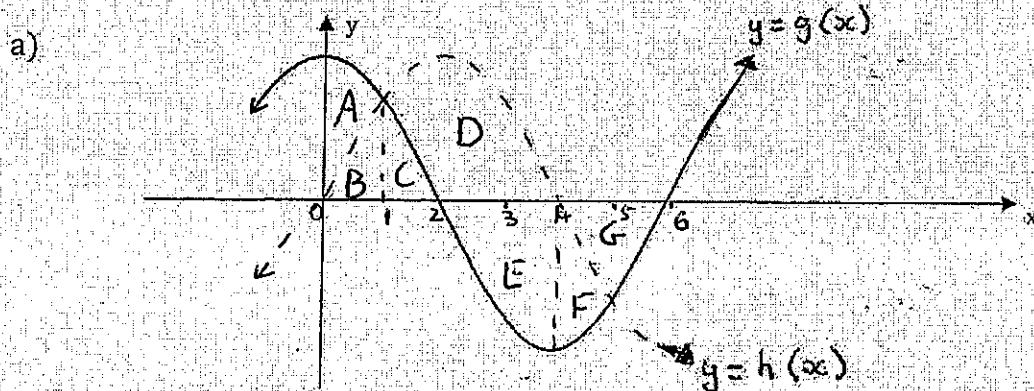
$$\int_1^3 x \sin x \, dx$$

- e) Find the volume formed when the area enclosed between  $y = x^2$  and  $y = 4x - x^2$  is rotated about the  $x$ -axis.

4

Question 14 - Start A New Booklet - (15 marks)

Marks



$A, B, C, D, E, F$  and  $G$  are the areas of the regions in which they are given.

Using these letters, write an expression for:

(i)  $\int_0^4 h(x) dx$

(ii)  $\int_1^4 g(x) dx$

1,2

b) Solve  $\tan 3\theta = 1$  for  $0 \leq \theta \leq 2\pi$

3

c) Find the equation of the parabola with vertex  $(-1, 1)$  and focus  $(-3, 1)$

3

d) (i) Differentiate

2

$$y = \log_e \left( \frac{x-1}{x+1} \right)$$

(ii) Hence, or otherwise, find

1

$$\int \frac{1}{x^2-1} dx$$

e) Given  $y = -4x - 20$  is the equation of a tangent to  $y = x^3 - 4x^2 - 7x + 10$  and  $x > 0$ , find the coordinates of the point of contact.

3

Question 15 – Start A New Booklet – (15 marks)

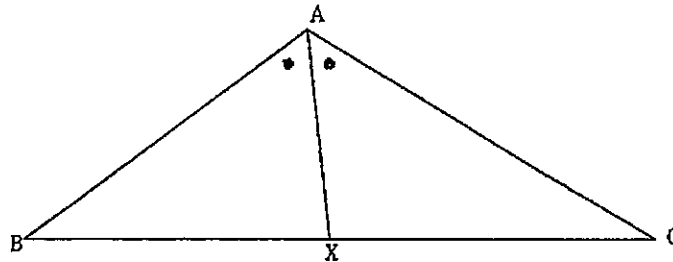
Marks

a) Simplify:

2

$$\frac{\sin^2 \theta}{\tan \theta \sin(90 - \theta)}$$

b)



Copy the diagram carefully onto your paper.

$X$  is a point on the side  $BC$  of  $\triangle ABC$  and  $AX$  bisects  $\angle BAC$ .

(i) Draw the line through  $X$  parallel to  $BA$  to meet  $AC$  at  $L$ .

This construction gives  $\frac{BX}{XC} = \frac{AL}{LC}$

1

(ii) Prove that  $\triangle ALX$  is isosceles.

2

(iii) Given that  $\triangle CAB \parallel \triangle CLX$  (Do not prove this) prove that  $\frac{BX}{XC} = \frac{AB}{AC}$

2

c) The equation of motion of a particle is  $x = te^{-t}$

where  $x$  is in centimetres  
 $t$  is in seconds.

(i) Find the time when the particle is at rest.

3

(ii) Find the equation of motion for acceleration and the acceleration when  $v = 0$ .

2

(iii) Find the time when acceleration is zero.

1

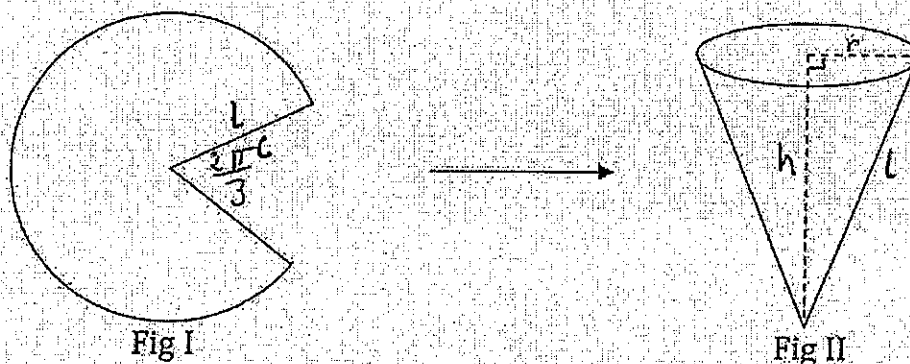
(iv) Using the answers from parts (i) to (iii) and other necessary information, sketch the displacement-time function  $x = te^{-t}$ . Show all important features clearly.

2

Question 16 – Start A New Booklet – (15 marks)

Mark

- a) An open cone, of radius  $r$  cm, and height,  $h$  cm is made from a sector of a circle. The area of the sector used is  $300 \text{ cm}^2$ .



- (i) Show from Figure I that slant height  $l$  is given by  $l^2 = \frac{450}{\pi}$  2

- (ii) Show from Figure II that  $h = \sqrt{l^2 - r^2}$  1

- (iii) Hence or otherwise show that the volume of the cone is given by 1

$$V = \frac{1}{3}r^2\sqrt{450\pi - \pi^2r^2}$$

- (iv) Show that  $\frac{dv}{dr} = \frac{300\pi r - \pi^2 r^3}{\sqrt{450\pi - \pi^2 r^2}}$  2

- (v) Find the value of  $r$  for the volume of the cone to be a maximum. 2

Question 16 – (cont'd)

Marks

b) Kando, the mathematical kangaroo always hops (i.e. jumps) according to mathematical rules. One day, Kando decides to go hopping according to the following rules:

- The length of odd number hops (1<sup>st</sup>, 3<sup>rd</sup>, 5<sup>th</sup> hop etc), in metres, is given by the arithmetic series  $t_n = 4 - (n - 1)$ , where  $n = 1, 3, 5, \dots$  is an odd number;
- The length of even number hops (2<sup>nd</sup>, 4<sup>th</sup>, 6<sup>th</sup> hop etc), in metres, is given by the geometric series  $T_N = \frac{192}{63} \left(\frac{1}{2}\right)^{\frac{N-2}{2}}$ , where  $N = 2, 4, 6, \dots$  is an even number;
- If the length of a hop is negative according to the relevant series, Kando hops the prescribed distance *backwards*.

- |                                                                            |   |
|----------------------------------------------------------------------------|---|
| (i) Write down the first term and common difference for the series $t_n$ . | 1 |
| (ii) Write down the first term and common ratio for the series $T_n$ .     | 1 |
| (iii) Find where Kando is relative to her starting point after 12 hops.    | 3 |
| (iv) Find the total distance travelled backwards in the first 16 hops.     | 2 |

End of Paper

# TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

Note  $\ln x = \log_e x, \quad x > 0$

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Multiple Choice

- |      |       |
|------|-------|
| 1. D | 6. C  |
| 2. C | 7. D  |
| 3. D | 8. C  |
| 4. A | 9. A  |
| 5. D | 10. B |

(iii)  $y = x \ln x$

$$y' = 1 \cdot \ln x + x \cdot \frac{1}{x}$$

$$= \ln x + 1$$

(e) (i)  $\int (3-2x)^4 dx = \frac{(3-2x)^5}{-2 \times 5} + C$

$$= -\frac{1}{10} (3-2x)^5 + C$$

Question 11

(a)  $\sqrt{\frac{4.83 \times 10.86}{17.83 - 5.92}} = 2.0986$

$$= 2.10$$

(ii)  $\int \frac{1}{\sqrt{x}} dx = \int x^{-\frac{1}{2}} dx$

$$= \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 2\sqrt{x} + C$$

(b)  $|2x-3| \leq 5$

$$-5 \leq 2x-3 \leq 5$$

$$-2 \leq 2x \leq 8$$

$$-1 \leq x \leq 4$$

(iii)  $\int \cos x^\circ dx = \int \cos \frac{\pi x}{180} dx$

$$= \frac{1}{\frac{\pi}{180}} \sin \frac{\pi x}{180} + C$$

$$= \frac{180}{\pi} \sin \frac{\pi x}{180} + C$$

(c)  $\log_a \sqrt{10} = \log_a 10^{\frac{1}{2}}$

$$= \frac{1}{2} \log_a 10$$

$$= \frac{1}{2} (\log_a 5 + \log_a 2)$$

$$= \frac{1}{2} (0.83 + 0.36)$$

$$= 0.595$$

(d) (i)  $y = \cos 7x$

$$y' = -7 \sin 7x$$

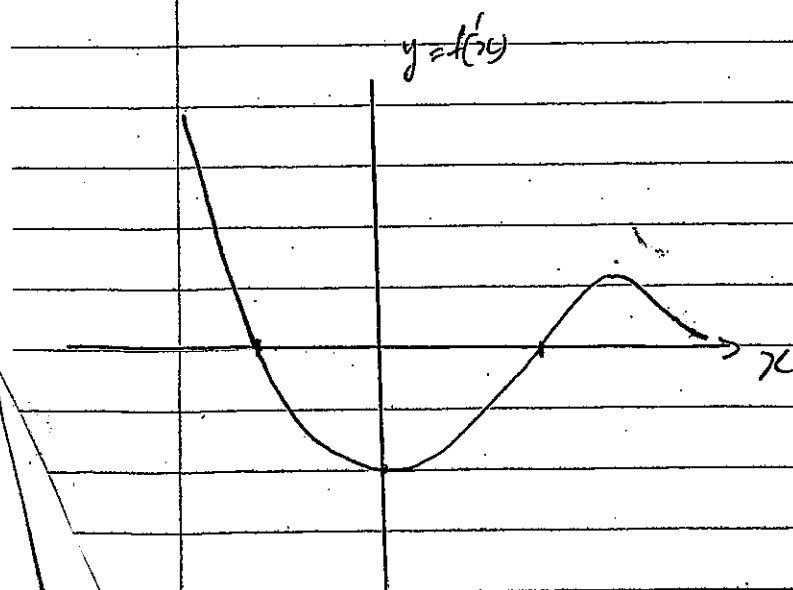
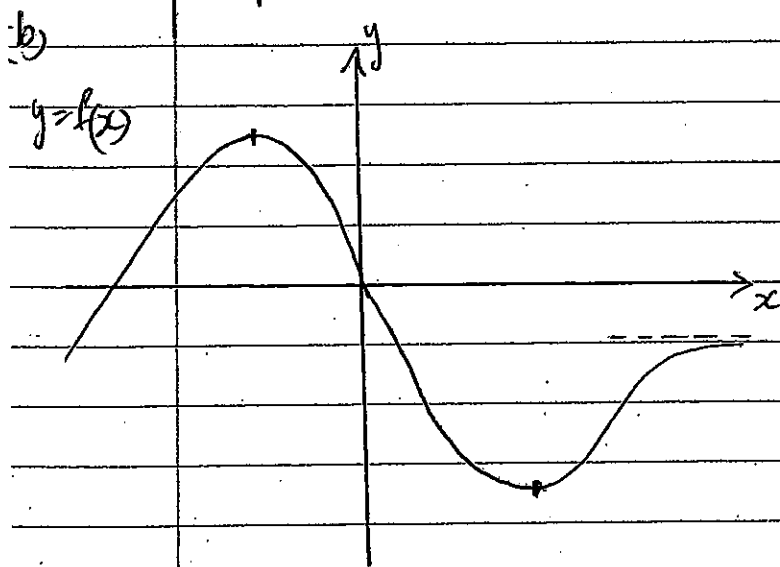
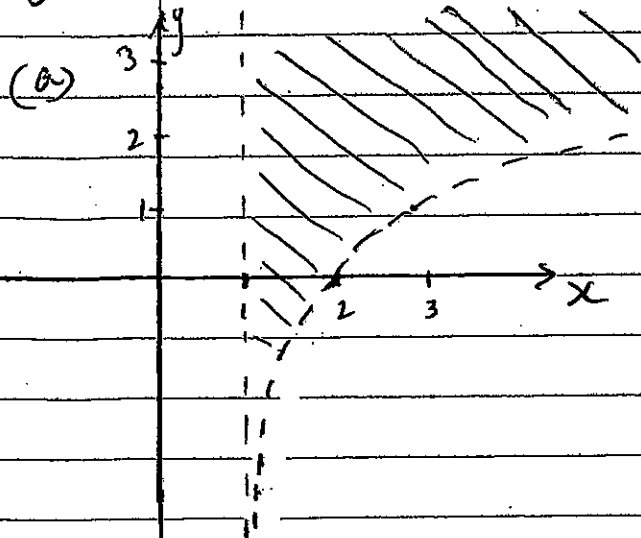
(ii)  $y = \sqrt{e^{2x} + 4}$

$$y = (e^{2x} + 4)^{\frac{1}{2}}$$

$$y' = \frac{1}{2} (e^{2x} + 4)^{-\frac{1}{2}} \times 2e^{2x}$$

$$= e^{2x} (e^{2x} + 4)^{-\frac{1}{2}}$$

## Question 12



(c)  $t=0$   $v=0$   $x=0$   $\therefore c=0$

$$a = 6t + 4$$

$$v = \int 6t + 4 \, dt$$

$$v = 3t^2 + 4t$$

$$x = \int 3t^2 + 4t \, dt$$

$$x = t^3 + 2t^2$$

(d) (i)  $d = \sqrt{(6-0)^2 + (3-7)^2}$   
 $= \sqrt{36+16}$   
 $= \sqrt{52}$

(ii)  $m = \frac{3-7}{6-0}$   
 $= -\frac{2}{3}$

(iii)  $m = \frac{3}{2}$   $(-2, 0)$

$$y - 0 = \frac{3}{2}(x + 2)$$

$$2y = 3x + 6$$

$$3x - 2y + 6 = 0$$

(iv)  $d = \frac{|2x_1 + 3y_1 - 21|}{\sqrt{2^2 + 3^2}}$   
 $= \frac{|2 \cdot -2 + 3 \cdot 0 - 21|}{\sqrt{13}}$   
 $= \frac{25}{\sqrt{13}}$

(v)  $c = (4, -4)$

(vi) Area = base  $\times$  height

$$= \sqrt{52} \times \frac{25}{\sqrt{13}}$$

$$= \sqrt{4} \times 25$$

$$= 50 \text{ sq units.}$$

# Question 13

a)  $n=20$   $r=0.5$

$$T_n = ar^{n-1}$$

$$20(0.5)^{n-1}$$

$$20(0.5)^{n-1} < 0.0001$$

$$(0.5)^{n-1} < 0.000005$$

$$(n-1) \log(0.5) < \log(0.000005)$$

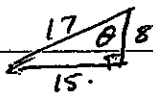
$$(n-1) > \frac{\log(0.000005)}{\log(0.5)}$$

$$n-1 > 17.6$$

$$n > 18.6$$

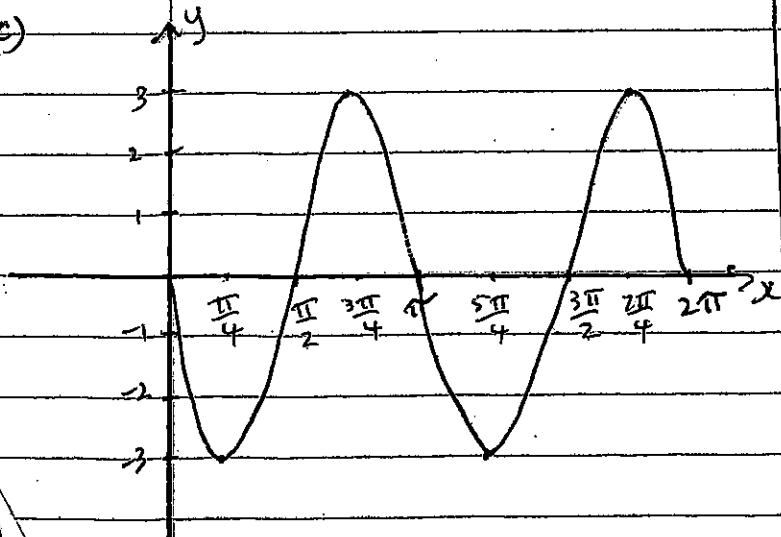
$\therefore$  19th term

b)  $\cos \theta = \frac{-8}{17}$



2nd Quadrant.

$$\therefore \sin \theta = \frac{15}{17}$$



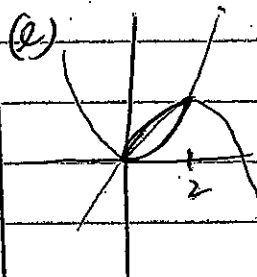
|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 1     | 1.5   | 2     | 2.5   | 3     |
| 0.841 | 1.496 | 1.819 | 1.496 | 0.423 |

$h=0.5$

$$\int_1^3 x \sin x dx \approx \frac{0.5}{3} \{ 0.841 + 4 \times 1.496 + 1.819 \}$$

$$+ \frac{0.5}{3} \{ 1.819 + 4 \times 1.496 + 0.423 \}$$

$$\approx 2.812$$



$$x^2 = 4x - x^2$$

$$0 = 4x - 2x^2$$

$$0 = 2x - x^2$$

$$0 = x(2-x)$$

$$x=0, 2$$

$$Vol = \pi \int_0^2 (4x - x^2)^2 dx - \pi \int_0^2 (x^2)^2 dx$$

$$= \pi \int_0^2 16x^2 - 8x^3 + x^4 dx - \pi \int_0^2 x^4 dx$$

$$= \pi \int_0^2 16x^2 - 8x^3 + x^4 - x^4 dx$$

$$= \pi \int_0^2 16x^2 - 8x^3 dx$$

$$= \pi \left[ \frac{16x^3}{3} - \frac{8x^4}{4} \right]_0^2$$

$$= \pi \left[ \left( \frac{16 \cdot 2^3}{3} - \frac{8 \cdot 2^4}{4} \right) - (0) \right]$$

$$= \pi \cdot \frac{32}{3}$$

$$= \frac{32\pi}{3}$$

# Question 14

(a) (i)  $\int_0^4 h(x) dx = B + C + D$

(ii)  $\int_1^{14} g(x) dx = C - E$

b)  $\tan 3\theta = 1$   $0 \leq \theta \leq 2\pi$

Q1 & Q3

$3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$

$\frac{17\pi}{4}, \frac{21\pi}{4}$

$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{21\pi}{12}$  (3, -20) is the point of contact

(c) Vertex = (-1, 1) focus = (-3, 1)

$a = -2$   $(y-k)^2 = 4a(x-h)$   
 $(y-1)^2 = 8(x+3)$

d) (i)  $y = \ln \frac{(x-1)}{(x+1)}$

$y = \ln(x-1) - \ln(x+1)$

$y' = \frac{1}{x-1} \cdot 1 - \frac{1}{x+1} \cdot 1$

$y' = \frac{1}{x-1} - \frac{1}{x+1}$   
 $= \frac{x+1 - (x-1)}{(x-1)(x+1)}$

(ii)  $\int \frac{1}{x^2-1} dx = \frac{1}{2} \int \frac{2}{x^2-1} dx$

$= \frac{1}{2} \ln \frac{(x-1)}{(x+1)} + C$

(b)

$y = x^3 - 4x^2 - 7x + 10$

$y' = 3x^2 - 8x - 7$

$m = -4$

$-4 = 3x^2 - 8x - 7$

$0 = 3x^2 - 8x - 3$

$(3x+1)(x-3)$

$x > 0 \therefore x = 3$

$y = 3^3 - 4 \cdot 3^2 - 7 \cdot 3 + 10$   
 $= -20$

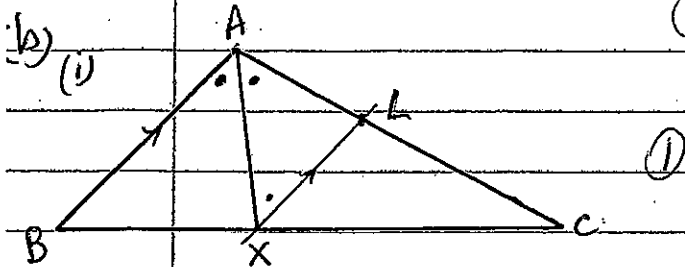
# Question 15

a)

$$\frac{\sin^2 \theta}{\tan \theta \sin(90-\theta)} = \frac{\sin^2 \theta}{\frac{\sin \theta}{\cos \theta} \cdot \cos \theta}$$

$$= \frac{\sin^2 \theta}{\sin \theta}$$

$$= \sin \theta \quad (2)$$



ii)  $\hat{BAX} = \hat{ALX}$  (alternate angles on parallel lines)

$\triangle ALX$  isosceles (base angles equal)

$$\therefore AL = LX \quad (2)$$

ii)  $\triangle CAB \parallel \triangle CLX$

$$\frac{AB}{LX} = \frac{AC}{LC}$$

$$\frac{AB}{AC} = \frac{LX}{LC}$$

$$\therefore \frac{AB}{AC} = \frac{AL}{LC} \quad (AL = LX)$$

$$\text{but } \frac{AL}{LC} = \frac{BX}{XC} \quad (\text{given})$$

$$\therefore \frac{AB}{AC} = \frac{BX}{XC} \quad (2)$$

(c)

$$x = te^{-t}$$

$$\dot{x} = 1 \cdot e^{-t} + t \cdot e^{-t}$$

$$= e^{-t}(1+t)$$

(i)

$$\dot{x} = 0$$

$$(1+t)e^{-t} = 0$$

$$1+t = 0$$

$$t = -1 \quad (3)$$

(ii)

$$\ddot{x} = -e^{-t}(1+t) + -1 \cdot e^{-t}$$

$$= e^{-t}(-1-t-1)$$

$$= (t-2)e^{-t}$$

$$t = 1$$

$$\ddot{x} = (1-2)e^{-1}$$

$$= -e^{-1}$$

$$= -\frac{1}{e} \quad (2)$$

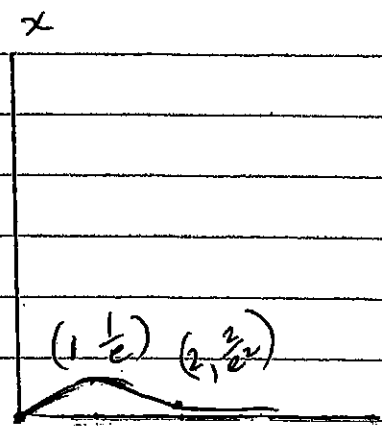
(iii)

$$\ddot{x} = 0$$

$$(t-2)e^{-t} = 0$$

$$t = 2 \quad (1)$$

(iv)



(2)

## Question 16

(i)  $A = \frac{1}{2} r^2 \theta$   $\theta = 2\pi - \frac{2\pi}{3}$

$\therefore 300 = \frac{1}{2} l^2 \frac{4\pi}{3}$   $= \frac{4\pi}{3}$

$900 = 2\pi l^2$

$\frac{450}{\pi} = l^2$

(ii)  $r^2 + h^2 = l^2$   
 $h^2 = l^2 - r^2$   
 $h = \sqrt{l^2 - r^2}$

(iii)  $V = \frac{1}{3} \pi r^2 h$

$= \frac{1}{3} \pi r^2 \sqrt{l^2 - r^2}$

$= \frac{1}{3} \pi r^2 \sqrt{\frac{450}{\pi} - r^2}$

$= \frac{1}{3} r^2 \sqrt{\pi^2 \left( \frac{450}{\pi} - r^2 \right)}$

$= \frac{1}{3} r^2 \sqrt{450\pi - \pi^2 r^2}$

(iv)  $V = \frac{1}{3} r^2 (450\pi - \pi^2 r^2)^{\frac{1}{2}}$

$V' = \frac{2}{3} r (450\pi - \pi^2 r^2)^{\frac{1}{2}} + \frac{1}{2} (450\pi - \pi^2 r^2)^{-\frac{1}{2}} \cdot -2\pi^2 r \cdot \frac{1}{3} r^2$

$= \frac{2}{3} r (450\pi - \pi^2 r^2)^{\frac{1}{2}} - \frac{\pi^2 r^3}{3} (450\pi - \pi^2 r^2)^{-\frac{1}{2}}$

$= \frac{2r(450\pi - \pi^2 r^2)}{3(450\pi - \pi^2 r^2)^{\frac{1}{2}}} - \frac{\pi^2 r^3}{3(450\pi - \pi^2 r^2)^{\frac{1}{2}}}$

$= \frac{300r\pi - \frac{2}{3}\pi^2 r^3 - \frac{1}{3}\pi^2 r^3}{(450\pi - \pi^2 r^2)^{\frac{1}{2}}}$

$= \frac{300\pi r - \pi^2 r^3}{\sqrt{450\pi - \pi^2 r^2}}$

(v)  $V' = 0$

$\frac{300\pi r - \pi^2 r^3}{\sqrt{450\pi - \pi^2 r^2}} = 0$

$300\pi r - \pi^2 r^3 = 0$

$300r - \pi r^3 = 0$

$r(300 - \pi r^2) = 0$

$r(\sqrt{300} + \sqrt{\pi}r)(\sqrt{300} - \sqrt{\pi}r) = 0$

$\therefore r = 0, -\sqrt{\frac{300}{\pi}}, \sqrt{\frac{300}{\pi}}$

$0, -\sqrt{\frac{300}{\pi}}$  are not valid solutions

$\therefore r = \sqrt{\frac{300}{\pi}}$

| r  | 9    | $\sqrt{\frac{300}{\pi}}$ | 10    |
|----|------|--------------------------|-------|
| V' | 37.8 | 0                        | -13.4 |

$\therefore r = \sqrt{\frac{300}{\pi}}$  gives max. volume.

# Question 16.

b) (i)  $T_n = 4 - (n-1)$

$T_1 = 4 \quad \therefore a = 4$

$T_2 = 2 \quad d = -2$

$T_3 = 0$

(ii)  $T_N = \frac{192}{63} \left(\frac{1}{2}\right)^{\frac{N-2}{2}}$

$T_1 = \frac{192}{63} \left(\frac{1}{2}\right)^0$

$= \frac{192}{63}$

$T_2 = \frac{192}{63} \left(\frac{1}{2}\right)^1$

$= \frac{192}{63} \cdot \frac{1}{2}$

$T_3 = \frac{192}{63} \left(\frac{1}{2}\right)^2$

$\therefore a = \frac{192}{63} \quad r = \frac{1}{2}$

(iii)  $S_n + S_N \quad n=6 \quad N=6$

$a=4 \quad d=-2$

$S_6 = \frac{6}{2} (2 \times 4 + 5 \times -2)$

$= 3(8-10)$

$= -6$

$a = \frac{192}{63} \quad r = \frac{1}{2} \quad S_n = \frac{a(1-r^n)}{1-r}$

$S_6 = \frac{192}{63} \cdot \left(1 - \frac{1}{2^6}\right)$

$= \frac{192}{63} \cdot \left(1 - \frac{1}{64}\right)$

$= \frac{384}{63} \left(1 - \frac{1}{64}\right)$

$= \frac{384}{63} \cdot \frac{63}{64}$

$= \frac{384}{64}$

$= 6$

$S_n + S_N$

$= -6 + 6$

$= 0$

$\therefore$  Kando is at the starting point.

(iv) Total distance backwards.

is  $-2 - 4 - 6 - 8 - 10 = -30$

$\therefore$  30 metres backwards.